A · Next-spike predictive foundation

AUDIO · TEXT · IMAGE · VIDEO · STATE / ACTION

Causal adapters · 16 slices + modality token

unified causal event streams

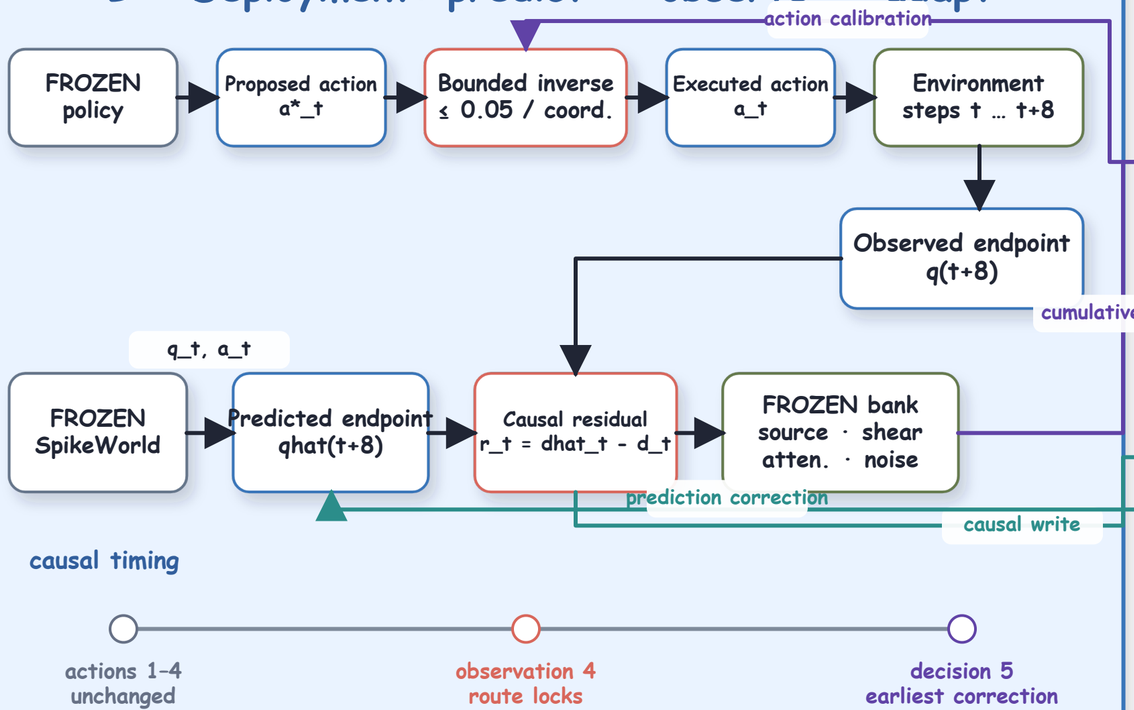
Shared sparse Spike Transformer
2 layers · width 96 · 4 heads
top-4 causal attention

predict the next event; read the present state

next-spike / next-slice · current semantics · image ↔ text binding · action-cond. next state

✓ selected checkpoint → FREEZE

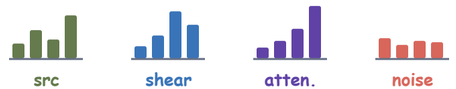
B · Deployment: predict → observe → adapt



ONLINE UPDATE USES OBSERVED TRANSITIONS ONLY

C · Bounded deployment state

Candidate-score state · 132 B
selects registered bounded inverse



Residual matrices · prediction state
refines future endpoints; does not choose action



28,848 B peak numeric state

no optimizer state · no replay · weights frozen

NEXT-SPIKE FOUNDATION

action next-state MSE ↓ 17.10%
semantic accuracy + 1.80 pp
binding R@1 + 1.81 pp

joint-formation study: separate checkpoints

29 KB CAUSAL DEPLOYMENT STATE

Prediction gain
+19.3%
registered stream

Tracking gain
+3.80%
calib. source: 0.000

parameters + fixed-probe logits unchanged in 9/9 runs

THE DEPLOYMENT OPERATING POINT

External state
28.8 KB · +19.3%

Pred.-path tuning
1.93 MB · +21.9%

19.3% versus 21.9% prediction gain
using 1/67 as much mutable numeric state